LABORATORY

Project No. 1

Financial Fraud Detection

Classification of Highly Imbalanced Data

Report No. 1

Theoretical aspects and data analysis

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1. Introduction

Financial fraud detection is one of the most important applications of Machine Learning (ML) in modern financial systems.

Banks, payment processors, and online platforms handle millions of transactions every day. Because of this massive scale, manual monitoring of every transaction is impossible. As a result, automated fraud detection systems are necessary to identify suspicious behavior and prevent financial losses.

In financial transaction datasets fraudulent operations usually represent a very small fraction of all transactions.

In many real-world cases the percentage of fraud cases is below 1%. This creates a serious challenge known as the **Class Imbalance (CI)** problem.

In a highly imbalanced dataset the majority class (legitimate transactions) dominates the dataset, while the minority class (fraudulent transactions) contains very few examples. Standard Machine Learning algorithms tend to focus on the majority class and may ignore patterns in the minority class.

Because of this issue, traditional evaluation metrics such as Accuracy (ACC) are not sufficient. A classifier that predicts every transaction as legitimate could achieve over 99% accuracy while completely failing to detect fraud.

Therefore fraud detection models must be evaluated using metrics that focus on minority class performance such as Recall (REC), Precision (PRE), F1-score (F1), Receiver Operating Characteristic Area Under Curve (ROC-AUC), and Precision–Recall Area Under Curve (PR-AUC).

This project focuses on understanding the fraud detection problem, analyzing a real dataset, and preparing the methodological foundation for implementing machine learning models capable of detecting fraudulent transactions.

2. Financial Fraud Detection in Machine Learning

Financial fraud detection refers to the process of identifying suspicious or illegal financial transactions.

Fraudulent activities may include credit card fraud, identity theft, transaction laundering, and other forms of financial abuse.

Machine Learning techniques have become widely used in fraud detection because they can identify complex patterns in large datasets. Traditional rule-based systems often fail to detect new fraud strategies, while machine learning models can adapt to new patterns.

Fraud detection systems typically perform binary classification where transactions are categorized as:

Class = 0 → Legitimate transaction

Class = 1 → Fraudulent transaction

The goal of a fraud detection model is to correctly identify fraudulent transactions while minimizing false alarms.

However, this task is challenging due to class imbalance and the evolving nature of fraud strategies.

3. Problem of Class Imbalance

Class imbalance occurs when the number of samples in one class is significantly larger than in another class.

In fraud detection datasets legitimate transactions may represent more than 99% of the data, while fraudulent transactions represent less than 1%.

This imbalance creates several challenges:

1) **Bias toward the majority class**. Machine learning algorithms learn patterns primarily from the majority class and may ignore minority examples.
2) **Misleading accuracy.** High accuracy does not necessarily mean good fraud detection performance.
3) **Poor minority class detection.** Models may classify almost all transactions as legitimate.

Because of this problem, special techniques are required to improve model performance on the minority class. These techniques include data balancing methods, cost-sensitive learning, and threshold tuning.

4. Dataset Description

For this project, we selected two datasets: the **Credit Card Fraud Detection Dataset (CCFD)** and the **PaySim Fraud Detection Dataset (PSFD)**, both from Kaggle.

The use of two datasets allows a broader analysis of fraud detection methods under different data conditions. The first dataset contains real anonymized credit card transactions, while the second one is a simulated mobile money transaction dataset.

4.1. Credit Card Fraud Detection Dataset (CCFD)

This dataset contains real credit card transactions made by European cardholders.

Dataset statistics:

- Total transactions: **284,807**
- Fraudulent transactions: **492**
- Legitimate transactions: **284,315**
- Fraud cases represent approximately **0.17%** of the dataset.

The dataset contains **30 variables** including:

- **Time** – time elapsed between transactions
- **Amount** – transaction amount
- **V1–V28** – anonymized numerical features obtained using Principal Component Analysis (PCA)
- **Class** – target variable representing fraud status

Due to privacy protection, the original transaction attributes were transformed using PCA. This transformation preserves the statistical structure of the data while removing sensitive information.

The dataset is highly imbalanced, which makes it suitable for testing fraud detection methods designed for rare-event classification problems.

4.2. PaySim Fraud Detection Dataset (PSFD)

The second dataset used in this project is the **PaySim dataset**. This dataset was generated using a simulator that models real transactional patterns in mobile payment systems.

Dataset statistics:

- Total transactions: **6,362,620**
- Fraudulent transactions: **8,213**
- Legitimate transactions: **6,354,407**
- Fraud cases represent approximately **0.129%** of the dataset.

The dataset contains **11 variables** including:

- **step** – a unit of time in hours
- **type** – type of transaction (for example CASH-IN, CASH-OUT, TRANSFER, PAYMENT, DEBIT)
- **amount** – transaction amount
- **nameOrig** – customer identifier of the sender
- **oldbalanceOrg** – sender's balance before the transaction
- **newbalanceOrig** – sender's balance after the transaction
- **nameDest** – customer identifier of the receiver
- **oldbalanceDest** – receiver's balance before the transaction
- **newbalanceDest** – receiver's balance after the transaction
- **isFraud** – target variable representing fraud status
- **isFlaggedFraud** – indicator for transactions automatically flagged by the system

Unlike the CCFD dataset, PaySim contains interpretable business-related features, which makes it easier to analyze transaction logic and suspicious behavior patterns.

At the same time, some variables such as customer identifiers and the flagged-fraud indicator require careful handling, because they may not be directly useful for modeling or may introduce methodological risks.

5. Exploratory Data Analysis (EDA)

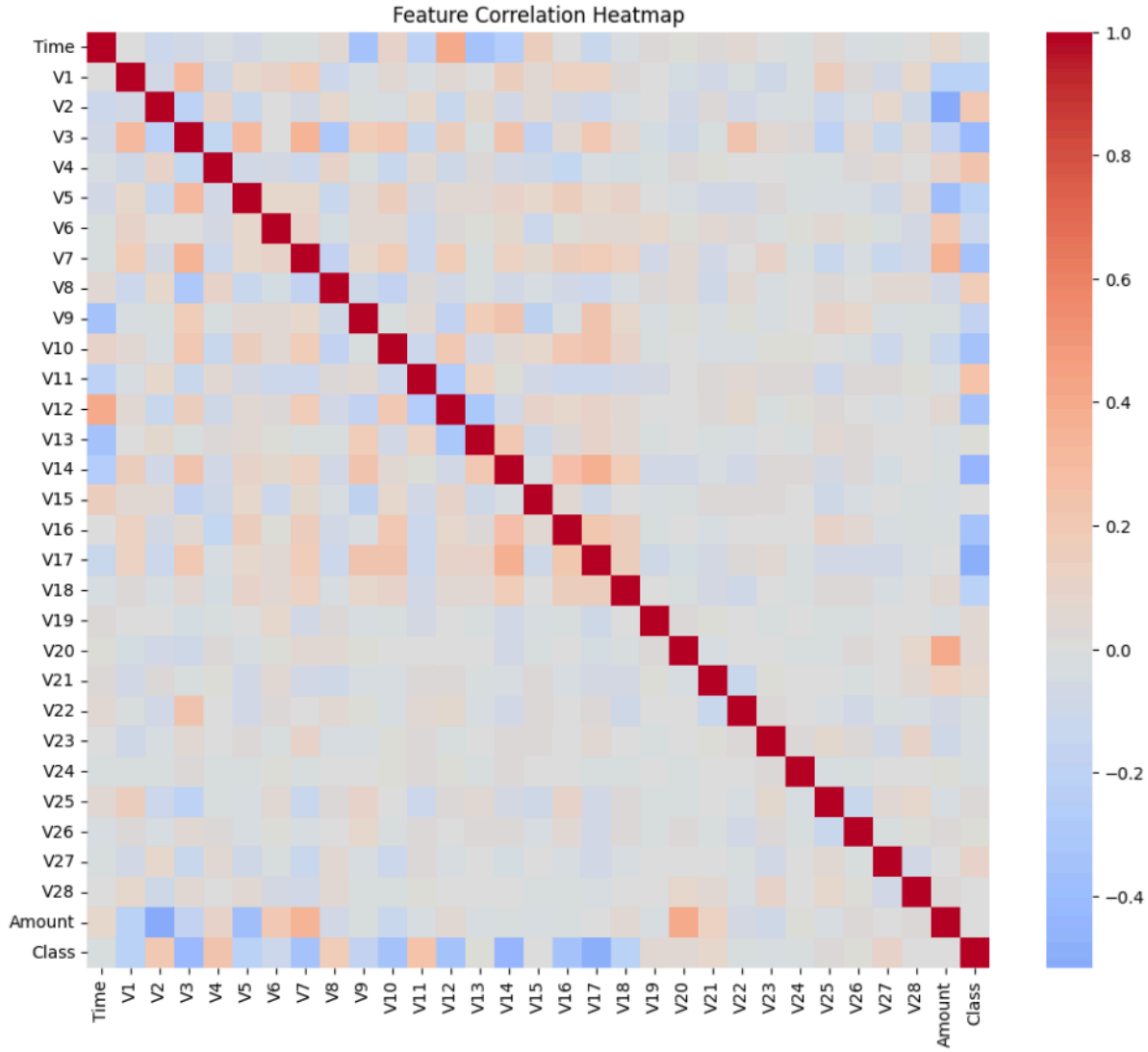
Exploratory Data Analysis (EDA) is an important step in understanding dataset characteristics.

EDA helps identify data distributions, correlations, anomalies, and potential preprocessing steps.

The following analyses will be performed:

- **Class Distribution Analysis**
  A bar chart will illustrate the extreme imbalance between legitimate and fraudulent transactions.
- **Transaction Amount Analysis**
  A histogram will compare the distribution of transaction amounts between legitimate and fraudulent transactions.

- **Correlation Analysis**
  A correlation heatmap will show relationships between variables in the dataset



- **Outlier Analysis**
  Fraud transactions often appear as outliers in certain feature distributions.
- **Missing Values Analysis**
  The dataset does not contain missing values, which simplifies preprocessing.

6. Risk of Data Leakage

Data Leakage (DL) is a common problem in machine learning experiments. It occurs when information from the test dataset influences the training process. This leads to overly optimistic model performance that does not reflect real-world behavior.

To prevent data leakage the following rules must be applied:

1) Data balancing techniques must be applied only to the Training Set (TRS)
2) The Test Set (TS) must remain untouched
3) Model evaluation must use unseen data

Ensuring correct experimental methodology is essential for producing reliable results.

7. Selected Machine Learning Models

Several Machine Learning models will be evaluated in the project:

- **Logistic Regression (LR)**
  A linear model commonly used as a baseline classifier. It is simple, interpretable, and computationally efficient.
- **Decision Tree (DT)**
  A tree-based model that splits the feature space using decision rules. It can capture nonlinear relationships between variables.
- **Random Forest (RF)**
  An ensemble learning method that combines multiple decision trees. Random Forest reduces overfitting and improves generalization performance.
- **Extreme Gradient Boosting (XGBoost)**
  A powerful gradient boosting algorithm widely used for structured data problems. It builds models sequentially, correcting errors from previous models.

Model	Role in study	Main strength	Main limitation
Logistic Regression	Baseline	Interpretable, easy to calibrate, fast	Limited for complex non-linear patterns
Decision Tree / Random Forest	Tree-based comparison	Captures interactions and non-linearity	Can still be biased toward majority class
XGBoost	Advanced ensemble	Often strong on tabular fraud data	Needs careful tuning and validation

8. Methods for Handling Class Imbalance

Several techniques will be used to address class imbalance:

- **Random Oversampling (ROS)**
  Duplicates minority class samples to increase their representation.
- **Random Undersampling (RUS)**
  Removes samples from the majority class to balance the dataset.
- **Synthetic Minority Oversampling Technique (SMOTE)**
  Generates synthetic minority class samples using interpolation between existing examples.
- **Adaptive Synthetic Sampling (ADASYN)**
  Focuses on generating new samples in regions where the minority class is more difficult to learn.

9. Evaluation Metrics

Evaluation metrics are crucial in fraud detection because accuracy alone is not informative.

The following metrics will be used:

- **Recall (REC)**
  Measures the proportion of fraudulent transactions correctly detected.
- **Precision (PRE)**
  Measures the proportion of predicted fraud transactions that are actually fraud.
- **F1-score (F1)**
  The harmonic mean of precision and recall.
- **ROC-AUC**
  Measures model performance across different classification thresholds.
- **PR-AUC**
  More informative metric for highly imbalanced datasets.

10. Cost-Sensitive Interpretation of the Fraud Detection Problem

In financial fraud detection, different classification errors have significantly different consequences. Traditional classification models assume that all errors have equal cost, but in real financial systems this assumption is unrealistic.

Therefore, fraud detection must be interpreted as a cost-sensitive classification problem.

When a classification model makes a prediction, four outcomes are possible:

Actual Class	**Predicted Legitimate**	**Predicted Fraud**
Legitimate	True Negative (TN)	False Positive (FP)
Fraud	False Negative (FN)	True Positive (TP)

Each type of error has a different operational impact:

- **False Negative (FN)**

A False Negative occurs when a fraudulent transaction is classified as legitimate. This is the most critical error because it allows fraudulent activity to proceed undetected.

Consequences include:

- direct financial loss for the bank or customer
- additional investigation costs
- reputational damage
- potential legal consequences

In practice, the financial loss caused by a single undetected fraud case may reach hundreds or even thousands of dollars.

- **False Positive (FP)**

A False Positive occurs when a legitimate transaction is classified as fraud. Although undesirable, this type of error is generally less costly.

Consequences include:

- temporary transaction blocking
- additional verification procedures
- customer inconvenience

However, the financial impact of a false alarm is usually limited to the operational cost of manual review.

1) **Cost Asymmetry**

Because the consequences of False Negative errors are significantly more severe than False Positive errors, fraud detection systems must prioritize minimizing FN errors.

This can be represented using a cost matrix:

Error Type	Example Cost
False Positive	$10–$20 (manual verification)
False Negative	$500–$2000 (fraud loss)

This large difference between costs creates cost asymmetry, which must be taken into account when designing machine learning models.

2) **Implications for Model Evaluation**

Due to the cost asymmetry in fraud detection:

a. maximizing accuracy is not sufficient
b. models should prioritize high Recall for the fraud class
c. evaluation metrics such as PR-AUC, Recall, and F1-score become more important

In many practical systems, it is preferable to generate more false alarms (FP) rather than miss fraudulent transactions (FN).

3) **Practical Perspective**

In real-world financial systems, fraud detection models are often integrated with human analysts. When the model detects suspicious transactions, they are flagged for manual review.

This hybrid approach allows the system to maintain high fraud detection rates while minimizing financial losses.

Therefore, cost-sensitive analysis plays a crucial role in the design and evaluation of fraud detection algorithms.

11. Research Hypotheses

The following hypotheses will be tested in the project:

- **H1**: Accuracy alone is not an appropriate metric for fraud detection.
- **H2**: Applying balancing techniques will significantly improve fraud Recall.
- **H3**: Ensemble models such as Random Forest and XGBoost will outperform simpler models.
- **H4**: Threshold tuning will improve fraud detection performance at the cost of reduced precision.

12. Conclusion

Fraud detection is a challenging machine learning problem due to extreme class imbalance and the high cost of misclassification errors.

This report presented the theoretical background of fraud detection, described the dataset(s) used in the study, and introduced the machine learning models and balancing methods that will be evaluated in future experiments.